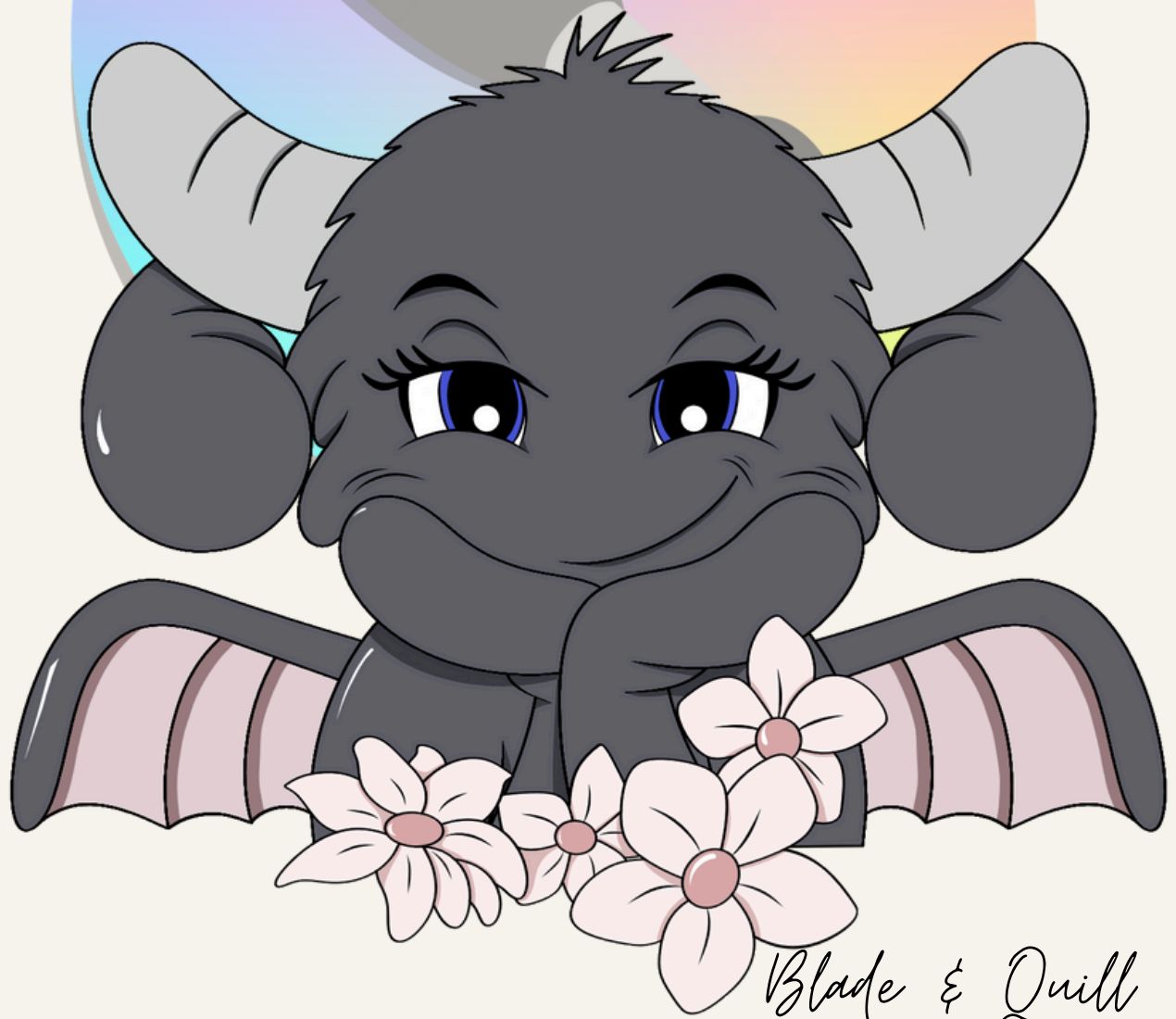


KRITA

FREE GUIDE



A GUIDE TO FIXING LAG

WHAT TO EXPECT FROM THIS GUIDE *Hello!*

This PDF file serves as a troubleshooting guide for users experiencing performance issues, specifically lag, within the digital painting software Krita.

It is structured around identifying **eight (8) common causes of lag** and providing actionable solutions for each.

The guide begins by emphasizing the crucial role of computer hardware, such as **RAM, CPU, GPU, and disk storage**, in Krita's performance.

Following this, it delves into **optimizing Krita's internal settings**, managing **autosave frequency**, reducing **layer count**, and **updating graphics drivers**, ultimately aiming to help you achieve a smoother and more responsive creative workflow.

Clickable links: 

If you see [blue](#), underlined text, it means you can click on it. When you do, it will either take you to another page or open a website.

BEFORE TO START

Important

Before starting, the most important prerequisite before implementing any optimization tips is to **first ensure that your computer's hardware meets the minimum or recommended specifications for running Krita effectively**, potentially requiring hardware updates or upgrades.

Check out Krita's recommendations on the next page. Also, you can refer to [page 14](#) to learn how to find information about your own computer. ✨



KRITA SYSTEM REQUIREMENTS

Krita recommends the following:

OPERATING SYSTEM : WINDOWS 8.1 OR HIGHER / MACOS 10.14 / LINUX

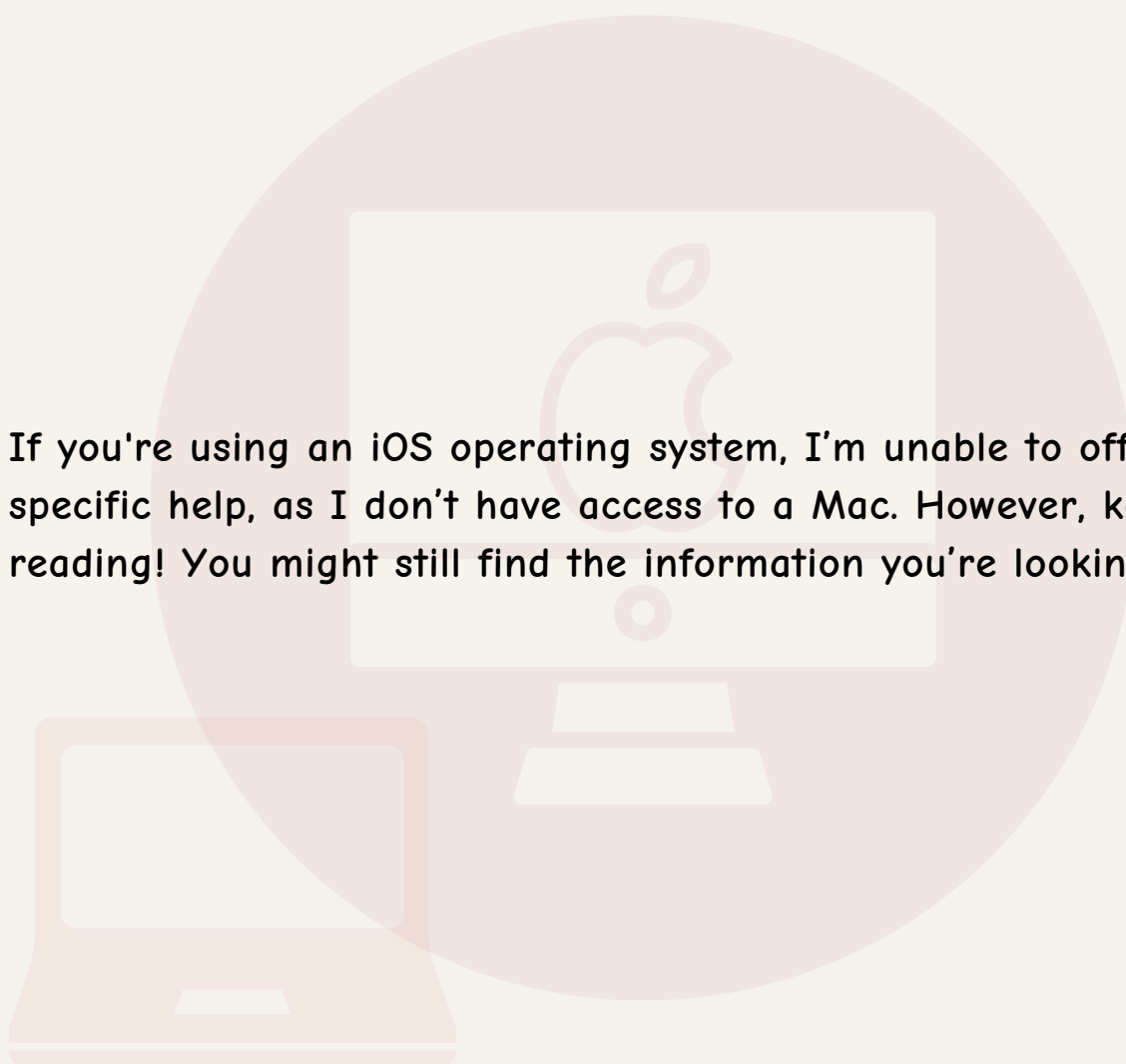
RAM : 4 GB AT MINIMUM REQUIRED. 16 GB RECOMMENDED.

GPU : OPENGL 3.0 OR HIGHER / DIRECT3D 11

GRAPHICS TABLET BRAND : ANY TABLET COMPATIBLE WITH YOUR OPERATING SYSTEM

DISCLAIMER

If you're using an iOS operating system, I'm unable to offer specific help, as I don't have access to a Mac. However, keep reading! You might still find the information you're looking for.



INTRODUCTION TO LAG

There are several reasons why lag can happen. I'll go over the 8 most common ones and show you how to fix them!

In Krita or any other painting software, lag can manifest itself in different ways:

1. A delay between your input and the on-screen result,
2. Choppy or broken lines,
3. The program freezes for brief periods,
4. Things move slowly,
5. Finally, whatever painting software you use, it crashes over and over.

If you are facing any of these problems, you've come to the right place. In this guide, I'll show you some solutions that I hope will help you boost Krita's performance, no more lag!

FIX #1 - HARDWARE CONSIDERATIONS

Krita's lag is most often caused by your computer's hardware setup. So, before we dive into software tweaks later in the guide, it's important to first understand how Krita uses your system's resources.

Please don't skip this part—it's essential to know how Krita depends on your **RAM**, **CPU**, **GPU**, and **disk storage** before you can effectively fix any performance issues.

I'll start with the **RAM**.

For Krita to properly work, you will need to have the proper amount of RAM. Therefore, you must make sure, first, that you have what it takes to run the software.

Ask yourself: "Is it possible that the source of my issues could be as simple as a hardware upgrade?"

FIX #1 - HARDWARE CONSIDERATIONS

RAM stands for **Random Access Memory**.

Think of RAM as your computer's short-term memory or workspace. It's where your computer keeps the things it's actively working on. The more RAM you have, the more things your computer can juggle at once without slowing down. It is important that you understand that RAM is volatile. It's only used when needed, meaning when you close a program, the information in it is usually cleared.

So, why is RAM so important?

Krita needs RAM to hold your project while you work. Bigger projects use more RAM. If Krita fills up the RAM, it starts using hard disk drive (HDD) space as temporary memory, and this makes Krita run much slower, causing lag, or even worse, the software may crash.

MY RECOMMENDATIONS

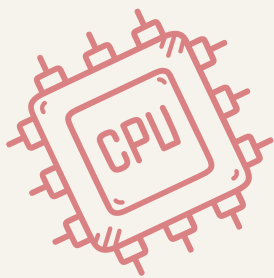
To reduce lag, aim for, at least **16 gibibytes (GB) of RAM**. The more RAM you have, the better Krita can handle large canvases, multiple layers, and a long undo history without slowing down.

It also helps a lot when working with animations.

So, if you're working on large projects or animations, I recommend getting **32 GB of RAM or more**. This will help prevent lag in the future.

I do an extensive amount of animations and paint large canvases. I have upgraded my computer RAM to **128GB!**

FIX #2 - CHECK YOUR CPU AND GPU COMPATIBILITY



Now, let's check your **CPU** and **GPU**.

Are the drivers updated, and does your GPU support Krita's OpenGL?



For those of you who are new to these technical terms, let me explain, very quickly, what they are.

CPU stands for **Central Processing Unit**: The CPU is like the "brain" of your computer. It is responsible for handling the core calculations within the software you're using. The faster your CPU, the quicker your computer can process information and get things done.

GPU stands for **Graphics Processing Unit**. Think of it as a specialized helper for the CPU, specifically focused on graphics and visuals. In Krita, its job is to accelerate the rendering process, especially if you have OpenGL enabled in Krita's settings.



I'll show you how to check which CPU and GPU you have, and how to upgrade them, on [page 33](#).



MY RECOMMENDATIONS

A faster CPU will improve overall responsiveness, especially with complex calculations and brush strokes. Multi-core CPUs are beneficial as Krita can utilize multiple cores for certain tasks.

Here are the minimum and optimal recommendations.

As of **April 2025**, the **minimum CPU and GPU recommendations** to run Krita smoothly are generally:

- CPU: 1.6 GHz Dual-core CPU.
- GPU: OpenGL 3.0 or higher / Direct3D 11 capable graphics card.

For **optimal performance**, especially when creating larger projects. here are the recommended CPU and GPU specifications:

- CPU: >4.0 GHz Quad-core CPU or higher.
- GPU: A modern dedicated GPU with at least 6 GB VRAM that supports OpenGL 3.2 or higher or Direct3D 12. NVIDIA and newer Intel GPUs generally provide the best results.

FIX #3 – MAYBE YOUR HDD IS OUTDATED OR TOO SLOW



Now, let's have a look at your disk storage, also referred to as a **Hard Disk Drive (HDD)**.

Your Hard Disk drive is your **permanent disk storage** for all your files.

Think of it as being your filing cabinet or your library!

It's important to have enough free space because when Krita exhausts your RAM, as mentioned earlier, it resorts to using the hard drive as temporary memory, severely impacting performance and, therefore, causing lag.

MY RECOMMENDATIONS

To avoid slow performance and lag, I strongly recommend upgrading to an **SSD**.

SSD stands for **Solid State Drive**. They are much faster and more responsive than regular hard drives. While they used to be expensive, prices in the last few years have become much more reasonable.

If you plan to create a lot of animations or work with multiple large files, and if your budget allows it, purchasing a **1 TeraByte (TB) SSD**, which is about 1,000 Gigabytes, would be a worthwhile investment for smoother performance.

However, if you cannot afford it or if you are not going to create any animations or large files, you may consider purchasing at least **250 or 500 Gigabytes**.



Your choice should be based on your needs, not just my recommendations. I'm simply offering these suggestions to help point you in the right direction.

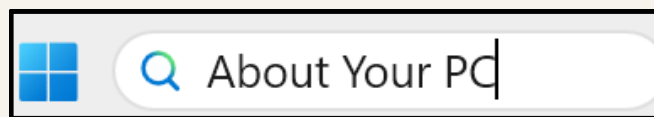
BONUS INFORMATION

It is important to know that Krita runs faster if it is installed on an SSD instead of your Hard Disk Drive.



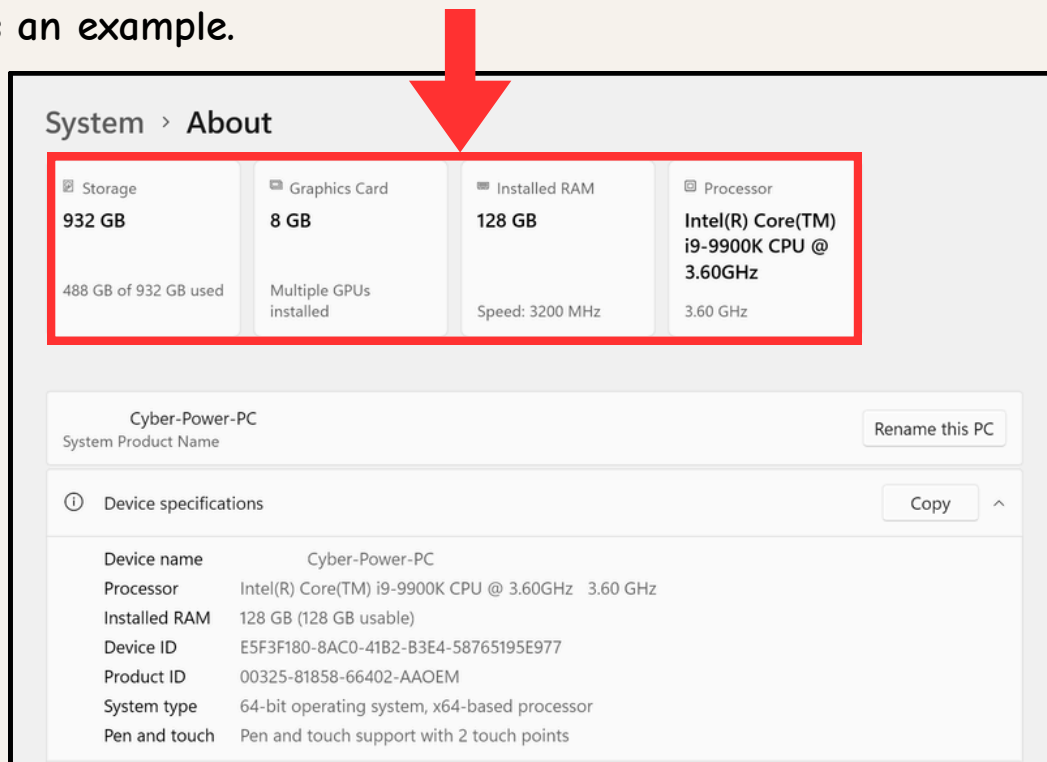
TIME TO CHECK YOUR COMPUTER!

In the search bar located at the bottom left of your screen, type “About your PC” and hit enter.



A new window will appear showing your disk storage (Storage), GPU (Graphics Card), RAM (Installed RAM), and CPU (Processor).

Below is an example.



Take a moment to compare what you have with what you actually need **before considering any upgrades** to your PC or laptop.

MOVING ON TO THE NEXT STEPS!

Alright, it is time now for me to show you how to optimize a few more things in Krita and on your computer.



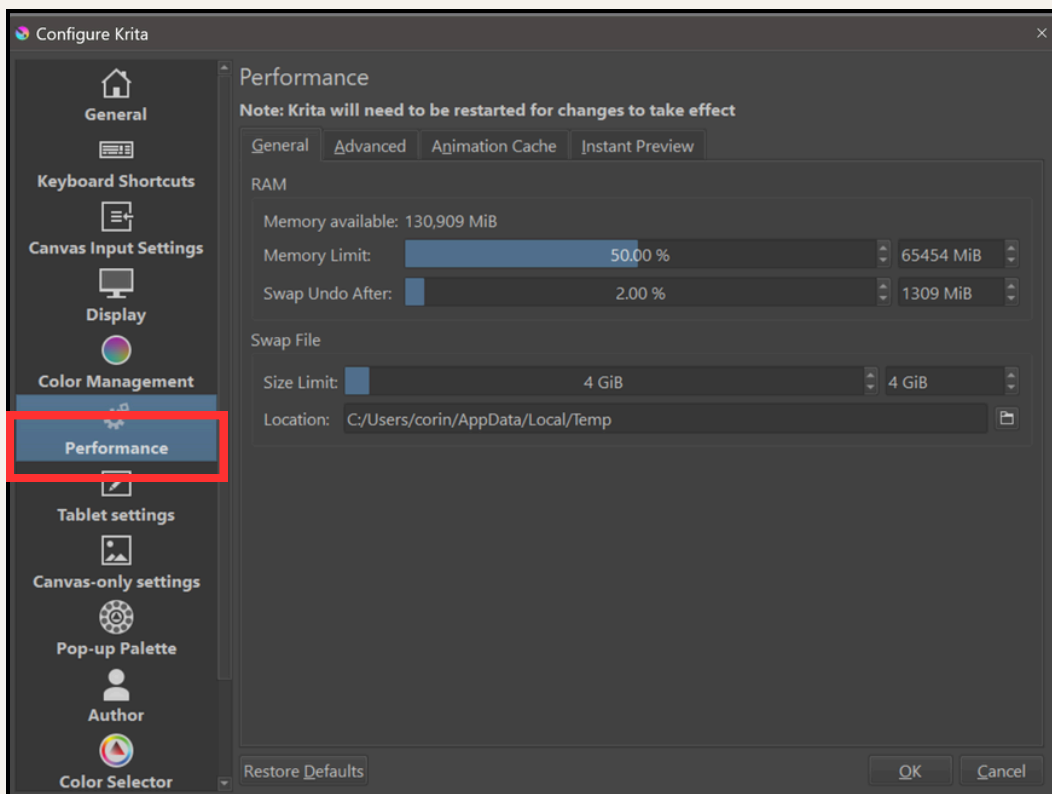
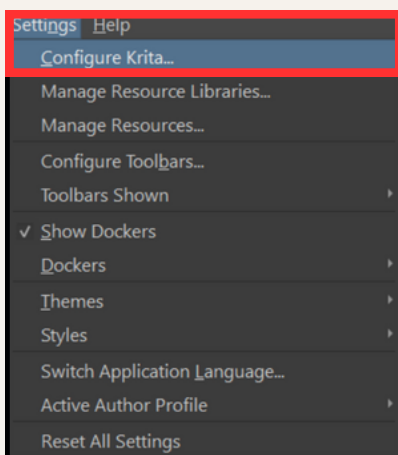
BUT BEFORE STARTING, ONLY FOLLOW THESE
INSTRUCTIONS AFTER YOU HAVE MADE SURE
THAT YOUR COMPUTER IS UP FOR IT.

MEANING THAT YOU HAVE UPDATED OR
UPGRADED YOUR PC OR LAPTOP FOR OPTIMAL
PERFORMANCE.

FIX #4 - OPTIMIZING KRITA SETTINGS

We are going to adjust Krita's internal settings to improve performance.

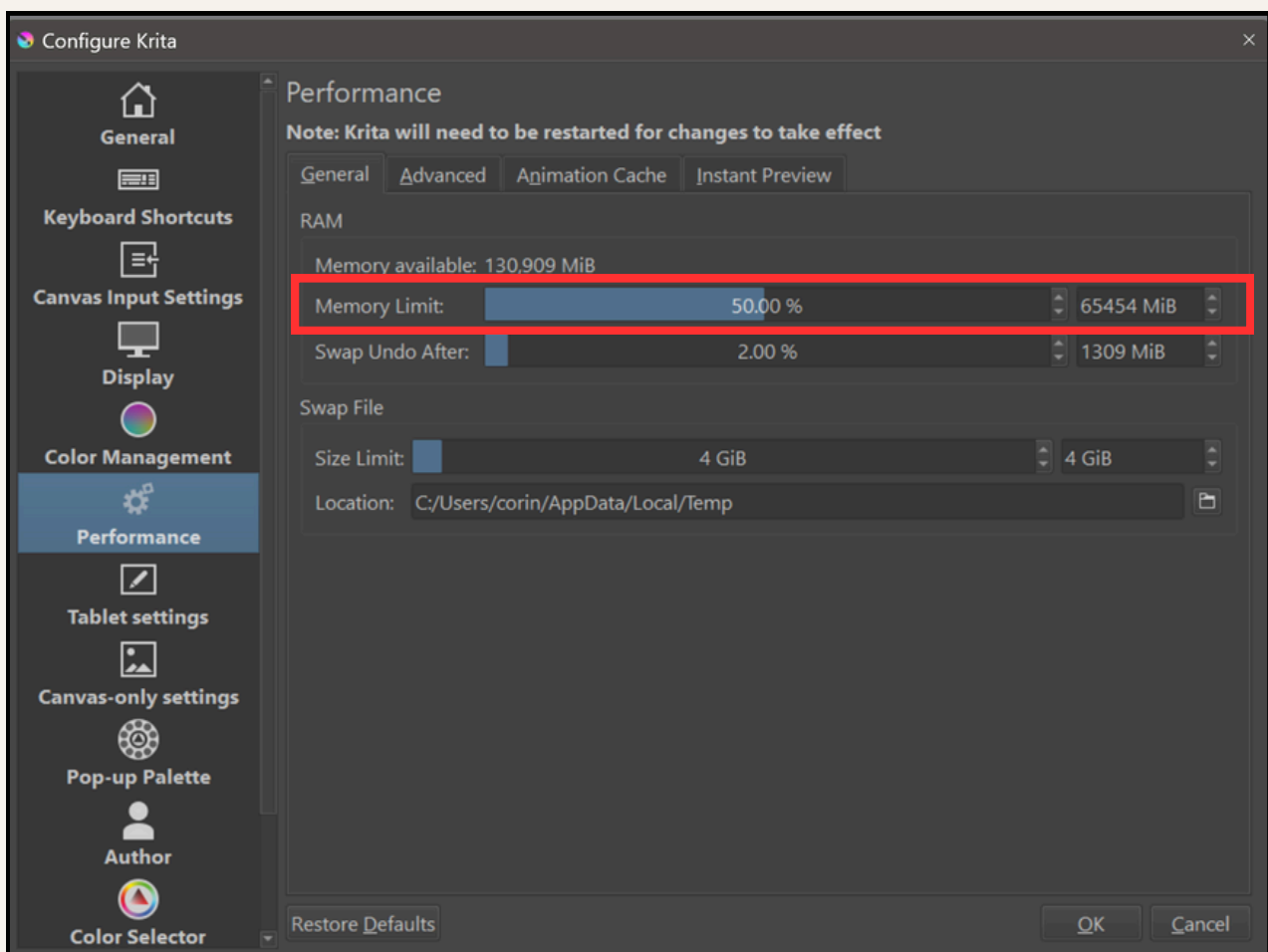
Go to Settings >Configure Krita >Performance.



FIX #4 - OPTIMIZING KRITA SETTINGS

Here is the **MEMORY LIMIT**. Increase Memory Limit (**only if you have enough RAM and based on your available RAM**).

When adjusting Krita's memory limit, it's generally a good idea to leave a significant portion of your system's RAM for the operating system and other applications.



MY RECOMMENDATIONS

Here are some guidelines I want you to consider:

- If your system has **only 4GB of RAM**: A safer approach is to set Krita's memory limit to around 30-40% (approximately 1.2GB to 1.6GB). This ensures the system has enough resources to run smoothly.
- If your system has **only 8GB of RAM**: You can likely set Krita's memory limit to around 40-60% (approximately 3.2GB to 4.8GB), providing a good balance between Krita's performance and overall system responsiveness.
- IF your system **has 16GB of RAM or more**: You have more flexibility. You can often experiment with 50-70% (8GB to 11GB+ for 16GB), leaving ample room to other tasks while allowing Krita to utilize more memory for larger projects.

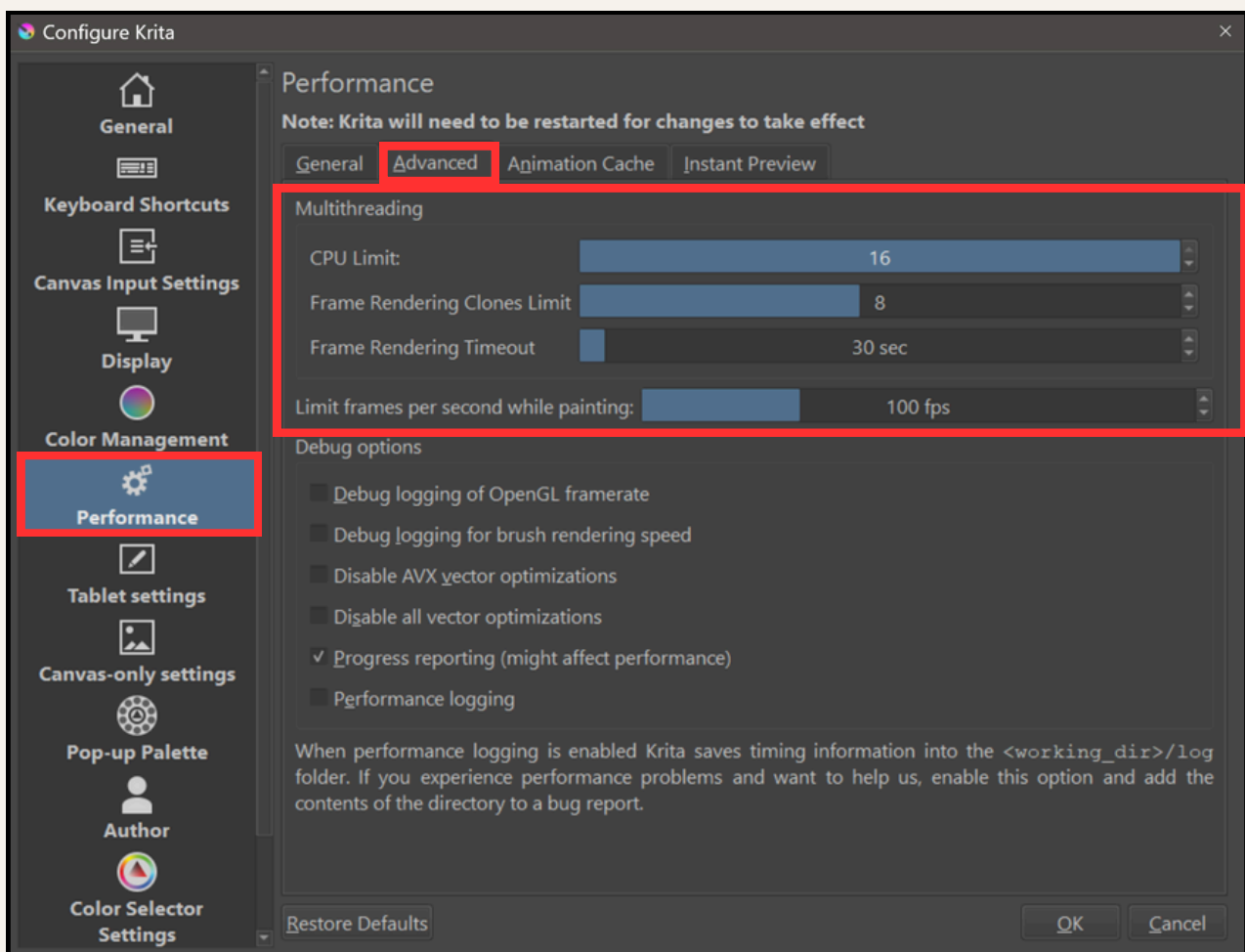


Before to continue, there are a few important things you need to know and make sure you follow:

1. Never allocate close to 100% of your RAM to Krita. It's a big NO-NO!
2. Monitor your system's performance after making changes. If you experiment slowdowns or instability outside of Krita, reduce the memory limit.
3. Finally, Restart Krita after adjusting the memory limit for the changes to take effect.

BONUS - MULTITHREADING

If you click on the “**Advanced**” tab, you’ll see a section called **Multithreading**.



BONUS - MULTITHREADING

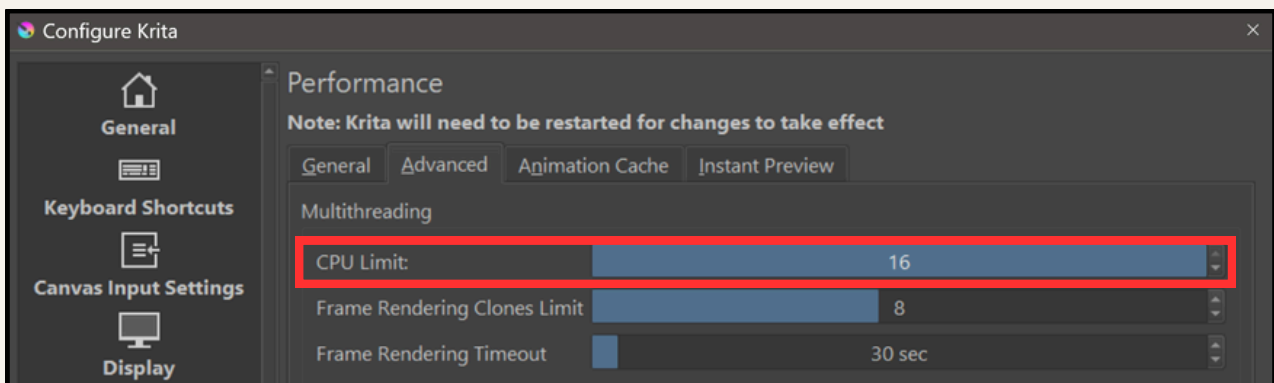
What it is: Multithreading is a technique that allows a program to divide its work into smaller, independent tasks (threads) that can be run simultaneously on multiple cores of your CPU. This can significantly speed up certain operations, especially those that involve a lot of calculations.



CPU LIMIT

Krita CPU Limit (Set to 16): This setting tells Krita the maximum number of CPU cores (or threads, in modern CPUs with hyperthreading) it's allowed to use for multithreaded tasks.

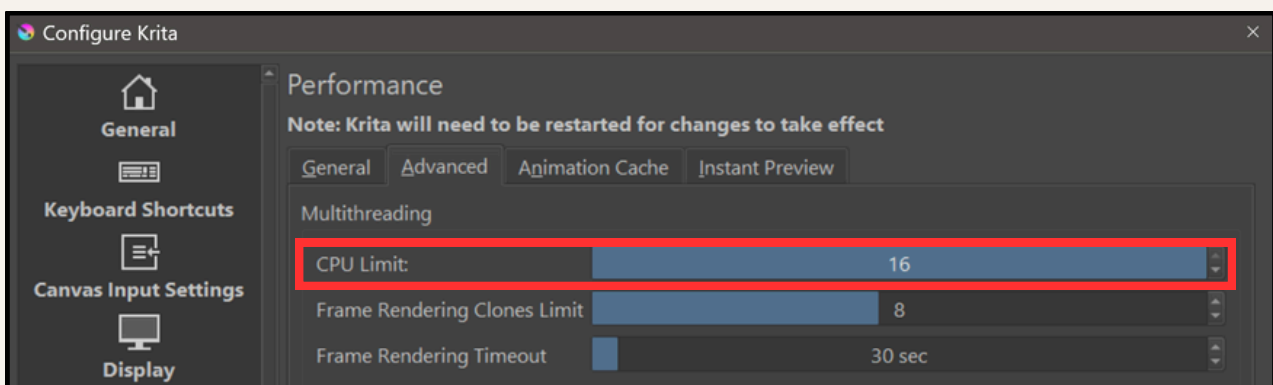
Maximum is 16: Since I have it set to the maximum of 16, Krita will try to utilize up to 16 of my CPU's processing units for tasks that can be parallelized.



CPU LIMIT (CONTINUES)

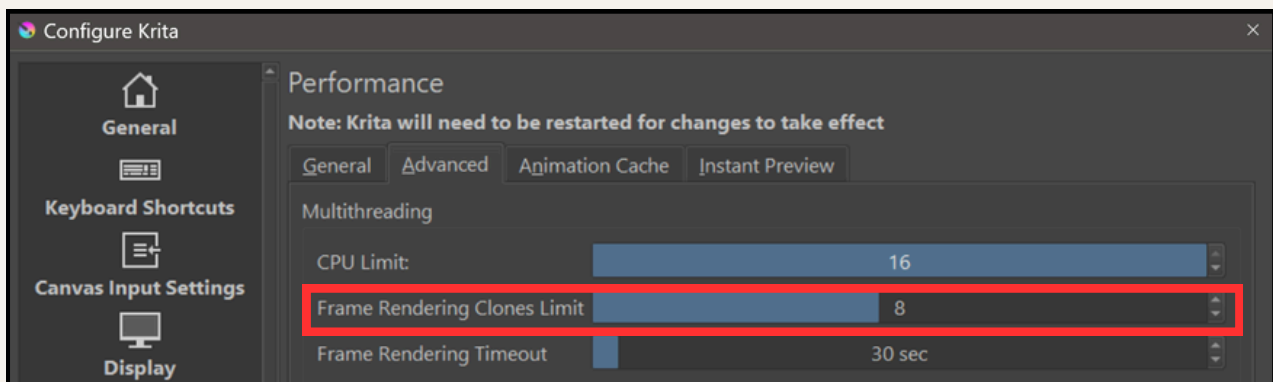
Benefit: The more cores Krita can use, the faster it can complete tasks like applying filters, rendering previews, and potentially some aspects of drawing (though brush performance is complex and not always linearly improved by more cores).

Note: The actual performance gain depends on how well Krita is optimized for multithreading for a specific task and the architecture of your CPU. More cores don't always mean a proportional increase in speed.



FRAME RENDERING CLONES LIMIT

What it is (in the context of animation rendering): When you render an animation to a sequence of individual image files, Krita can use multiple "clones" of your image in memory to process different frames (or parts of frames) simultaneously. This is another way to speed up the rendering process by utilizing multiple CPU cores.



FRAME RENDERING CLONES LIMIT

Frame Rendering Clones Limit (Set to 8): This setting limits the number of image clones Krita will create in memory during animation rendering.

Maximum is 16: This suggests Krita can potentially create up to 16 simultaneous rendering processes.

Setting to 8: I've set it to 8, meaning Krita will use up to 8 copies of my animation frame to render in parallel.

Benefit: For complex animations or high resolutions, using multiple clones can significantly reduce the overall rendering time.

Considerations: Each clone consumes a significant amount of RAM, especially for large or high-resolution animations. Setting this value too high, especially if you have limited RAM, could lead to memory exhaustion and system slowdowns or crashes. A balance needs to be struck between rendering speed and available RAM.



FRAME RENDERING TIMEOUT

What it is: During animation rendering, if Krita gets stuck on a particular frame for an extended period, it might indicate a problem.

The "**Frame Rendering Timeout**" setting tells Krita how long (in seconds) it should wait for a single frame to render before it considers the rendering process to have failed or hung.



FRAME RENDERING TIMEOUT

Frame Rendering Timeout (Set to 30 sec): I have set mine to 30 seconds. If Krita spends 30 seconds trying to render a single frame without completing it, it might stop the rendering process and potentially display an error.

Maximum is 600 sec (10 minutes): This shows the longest Krita will wait for a single frame to render before timing out.

Benefit: This setting helps prevent Krita from getting completely stuck on a problematic frame during a long animation render, allowing you to identify potential issues.

Considerations:

Complex Frames: Very complex frames with many layers, filters, or high resolutions might genuinely take a longer time to render. If you frequently encounter timeouts, you might consider increasing this value.

Underlying Issues: Frequent timeouts can also indicate a more serious problem with your hardware, software configuration, or the complexity of your animation project. Simply increasing the timeout might mask the underlying issue.

MUTITHREADING - SUMMARY

If you look at my settings, this is what they allow me to do.



CPU Limit (16): I am allowing Krita to use as many CPU cores as my system has for general multithreaded tasks.

Frame Rendering Clones Limit (8): I am telling Krita to use up to 8 parallel processes to speed up animation rendering, which can significantly reduce rendering time but consumes more RAM. However, my computer has 128GB of RAM. So, I am good to go!

Frame Rendering Timeout (30 sec): I am setting a 30-second limit for how long Krita will try to render a single animation frame before potentially giving up to avoid getting stuck.

MUTITHREADING - CONCLUSION

These settings are all about balancing performance and resource usage. My current settings seem like a reasonable starting point, especially since I have a multi-core CPU and sufficient RAM.

However, you might need to adjust the "Frame Rendering Clones Limit" and "Frame Rendering Timeout" based on the complexity and resolution of your animations and the amount of RAM available on your system.

If you need help, I would recommend you reach out to KRITA at the FORUM and ask for help. Just click on the link below, and it will take you to their support and advice page.

<https://krita-artists.org/c/support/6>



FIX #5 - OPTIMIZE GRAPHICS SETTINGS

Time to optimize your GPU,

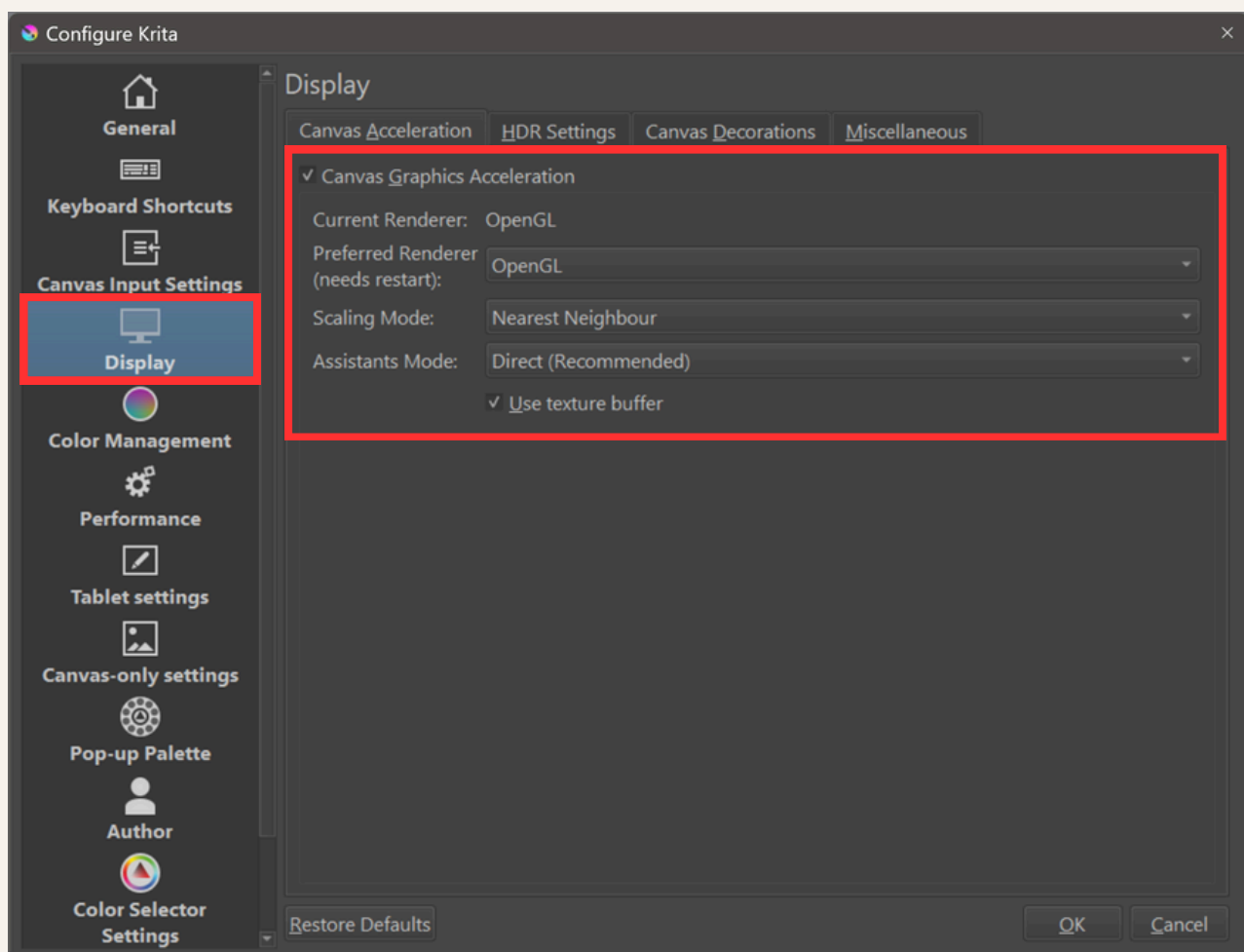
Go to Settings > Configure Krita > and select Display.

Make sure that the preferred **Renderer** is set to **OpenGL**.

Set **Scaling Mode** to **Nearest Neighbor**.

Leave the **Assistant Mode** to **Direct (recommended)**.

Click **Ok** to apply the changes!



FIX #5 - OPTIMIZE GRAPHICS SETTINGS



Once again, these are just recommendations. Depending on your computer and any upgrades you've made, you might need different settings.

Don't hesitate to experiment and find what works best for you.

For instance, Scaling Mode set to Nearest Neighbor may not be the best option for your projects or your computer settings.

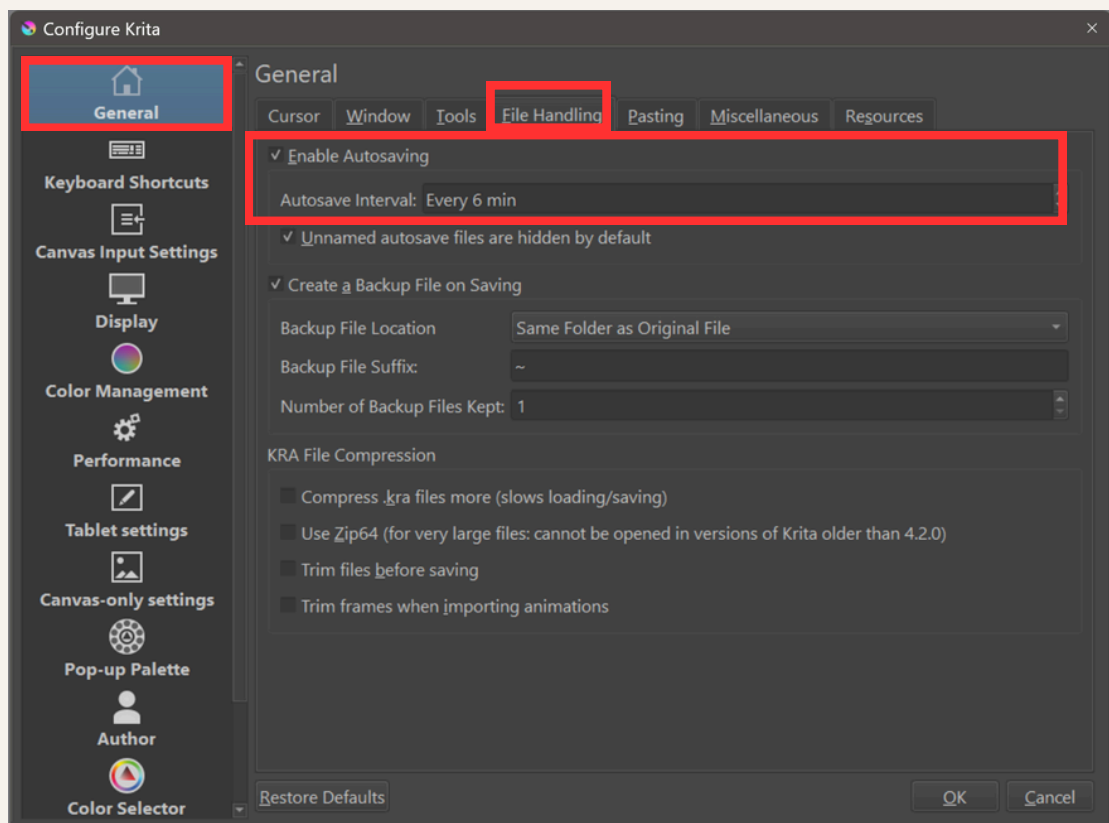
FIX #6 - MANAGE AUTOSAVE SETTINGS

I always recommend turning on **autosave** to help prevent accidental loss of your work. However, if autosave is set to run every minute, it can sometimes cause brief lag.

To avoid this:

Go to Settings > Configure Krita > in the General area > File Handling,

Set the autosave interval to every **6 minutes** instead of every 1 minute. This will help reduce interruptions caused by frequent saving.



FIX #7- REDUCE LAYERS & MERGE WHEN POSSIBLE

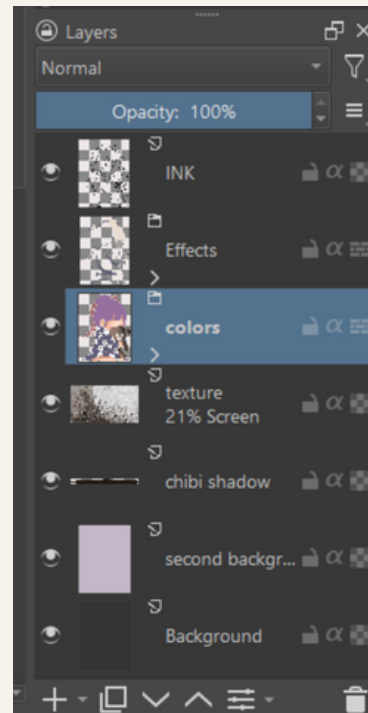
Having too many layers can make your file very large, which can slow things down.

Try **grouping related layers** to keep things organized and help reduce processing time.



In this example, I have 3 groups:

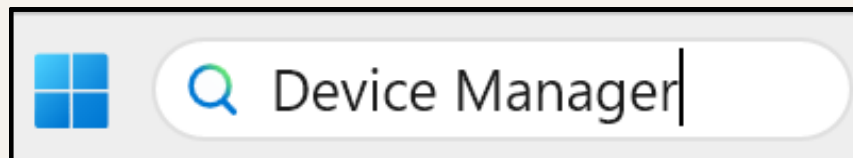
- Effects,
- Colors,
- and Color.



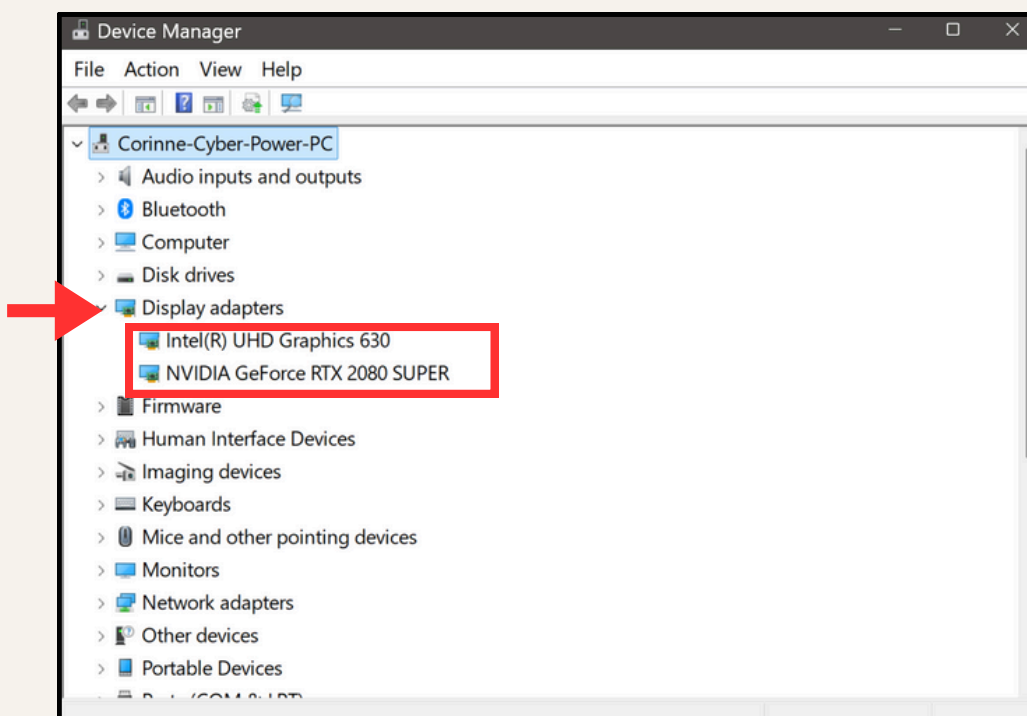
Finally, and if possible, consider **merging some layers**. The smaller the file, the less lag you'll experience.

FIX #8 - UPDATE GRAPHICS DRIVERS

We are almost done. Consider updating your **graphics drivers**. Just like earlier, get to the **search bar** located at the bottom left of your monitor, but this time, type **Device Manager**.

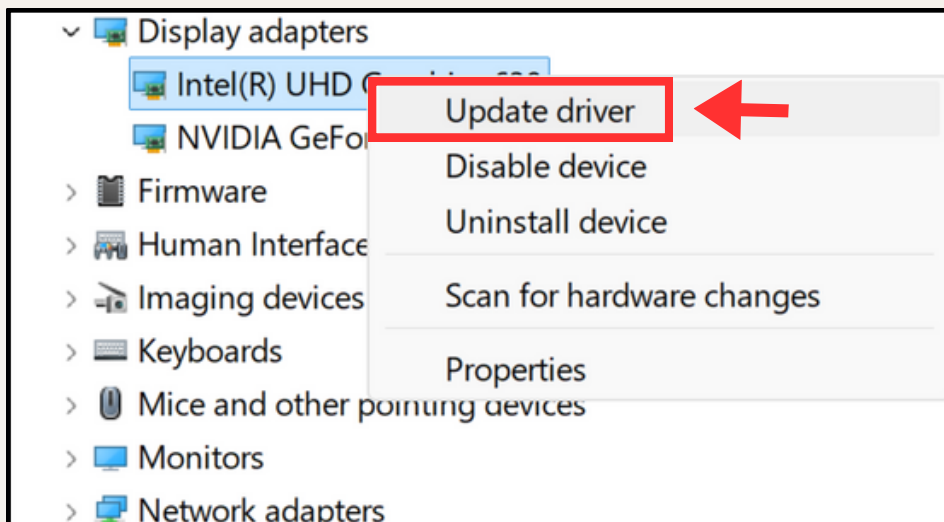


The window Device Manager will appear on your monitor. Expand '**Display adapters**' by clicking the arrow right next to it.



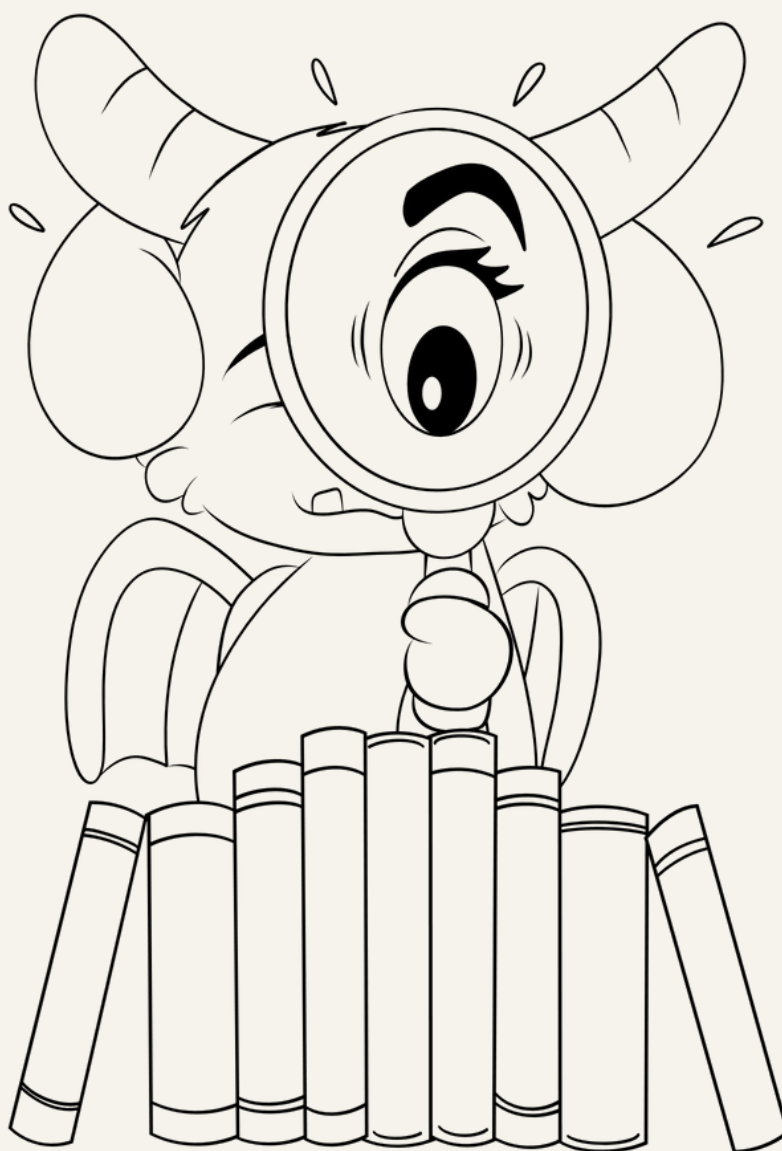
FIX #8 - UPDATE GRAPHICS DRIVERS

The last thing you need to do is right-click on all the GPUs you may have and select 'Update Driver'



GLOSSARY OF KEY TERMS

Must know!



GLOSSARY OF KEY TERMS

- **Lag:** In the context of digital painting, lag refers to a noticeable delay or slowness in the software's response to user input, resulting in a disconnect between the action (e.g., moving a pen) and the on-screen result.
- **RAM (Random Access Memory):** A type of computer memory that stores data that is actively being used so that it can be quickly accessed by the CPU. It is volatile memory, meaning data is lost when the power is turned off.
- **CPU (Central Processing Unit):** The main processing unit of a computer, responsible for executing instructions and performing calculations. Often referred to as the "brain" of the computer.
- **GPU (Graphics Processing Unit):** A specialized electronic circuit designed to rapidly manipulate and alter memory to accelerate the creation of images, videos, and other visual content.
- **OpenGL:** A cross-language, cross-platform API (Application Programming Interface) for rendering 2D and 3D vector graphics. Krita can utilize OpenGL for accelerated rendering.

GLOSSARY OF KEY TERMS

- **HDD (Hard Disk Drive):** A traditional type of computer storage that uses magnetic platters to store and retrieve digital data. HDDs are generally slower than SSDs.
- **SSD (Solid State Drive):** A type of computer storage that uses solid-state flash memory to store data. SSDs are significantly faster and more durable than HDDs.
- **Driver (Graphics Driver):** Software that allows the operating system and applications to interact with the computer's graphics card. Keeping drivers updated is important for performance and stability.
- **Autosave:** A software feature that automatically saves the user's work at specified intervals to prevent data loss in case of crashes or other issues.
- **Multithreading:** A technique that allows a program to execute multiple parts (threads) of a task concurrently, potentially improving performance on multi-core processors.
- **Scaling Mode (Nearest Neighbor):** An algorithm used when resizing images. Nearest Neighbor is a simple method that preserves sharp edges but can result in a pixelated appearance when scaling up.

CONCLUSION

These are the 8 most common reasons why Krita might lag.

Of course, there are other possible causes, but fixing these eight issues should noticeably improve your software's performance.

Lastly, I want to stress that you don't have to make any changes if you're not comfortable. If adjusting your computer or Krita settings feels overwhelming, it's perfectly okay to leave things as they are. Meaning, just keep using Krita's default settings.

If you really want to upgrade your computer, maybe a friend or family member can help you out.

This is the end of the guide. I hope you will find it helpful.

THANK YOU!
Bye!

*Thank
You*

THANK YOU, AGAIN, FOR FOLLOWING MY LITTLE CHANNEL.
IT IS TRULY AN HONOR TO HELP YOU ALL.

CORINNE.